

AI Ecosystem for Fake Job Scam Detection and Fraud-Aware Job Recommendation

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstract

Online employment fraud has become a severe and growing threat across global digital recruitment platforms, with Sri Lanka experiencing a sharp rise in fake job scam incidents that disproportionately target Sinhala and Tamil-speaking job seekers. All existing automated detection systems operate exclusively on English-language data and produce only binary real-or-fake verdicts with no explanation or risk guidance, leaving millions of multilingual users entirely unprotected.

This research proposes a multilingual AI-powered fake job post detection system supporting English, Sinhala, and Tamil. The system accepts both typed text and image-based job posts — including WhatsApp screenshots and scanned flyers — extracting text via Tesseract OCR before analysis. A cross-lingual transformer model, XLM-RoBERTa, is fine-tuned on a newly constructed trilingual dataset scraped directly from Sri Lankan job portals — topjobs.lk and ikman.lk — and manually labelled as fake or legitimate by trained annotators. A novel three-tier Suspicion Scoring Mechanism maps each posting to a Low, Medium, or High risk level using a weighted composite of model probability, heuristic keyword signals, structural completeness, and email domain type. SHAP and LIME explainability frameworks generate human-readable per-prediction explanations in the user's language. The complete system is deployed as a React Native mobile application backed by a FastAPI REST API on AWS infrastructure.

The expected outcomes include a validated multilingual detection model achieving a minimum F1-score of 90% across all three languages, the first trilingual fake job dataset for South Asia built from real Sri Lankan job portal data, and a production-ready mobile application evaluated through a user study of at least 50 participants targeting a System Usability Scale score above 70. Socially, the system protects vulnerable job seekers from financial harm and advances SDG 8 and SDG 16. Commercially, the system offers a freemium mobile product and a B2B API licensing model targeting Sri Lankan job portals.

Keywords: Fake Job Detection, XLM-RoBERTa, Multilingual NLP, Suspicion Scoring, Explainable AI

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List of Abbreviations

Table 1 List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
BLEU	Bilingual Evaluation Understudy
B2B	Business to Business
CCID	Computer Crimes Investigation Division
CER	Character Error Rate
CI/CD	Continuous Integration / Continuous Deployment
CoA	Centroid of Area
DSR	Design Science Research
ECS	Elastic Container Service
ELFJP	Elevated Learning strategy for Fake Job Post identification
EMSCAD	Employment Scam Arabic Dataset
ENN	Edited Nearest Neighbours
F1	F1-Score (Harmonic Mean of Precision and Recall)
FastAPI	Fast Application Programming Interface (Python Framework)
FLIS	Fuzzy Logic Inference System
GAN	Generative Adversarial Network
GST	Goods and Services Tax
HTTPS	Hypertext Transfer Protocol Secure
IC3	Internet Crime Complaint Center
iOS	iPhone Operating System
IT	Information Technology
JWT	JSON Web Token
K-NN	K-Nearest Neighbours
LKR	Sri Lankan Rupee
LIME	Local Interpretable Model-Agnostic Explanations

LLM	Large Language Model
MLP	Multi-Layer Perceptron
ML	Machine Learning
MongoDB	MongoDB (Document-oriented NoSQL Database)
NLP	Natural Language Processing
NoSQL	Non-relational Structured Query Language
OCR	Optical Character Recognition
POS	Part-of-Speech
REST	Representational State Transfer
ROC-AUC	Receiver Operating Characteristic — Area Under the Curve
S3	Simple Storage Service
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
SMOTE	Synthetic Minority Oversampling Technique
SMOTE-ENN	Synthetic Minority Oversampling Technique — Edited Nearest Neighbours
SUS	System Usability Scale
SVM	Support Vector Machine
TF-IDF	Term Frequency — Inverse Document Frequency
TLS	Transport Layer Security
UI	User Interface
URL	Uniform Resource Locator
USD	United States Dollar
XAI	Explainable Artificial Intelligence
XLM-RoBERTa	Cross-lingual Language Model — Robustly Optimised BERT Pre-training Approach

List of Appendices

Appendices - A	Survey Instrument
Appendices - B	Detailed Diagram
Appendices – C	UI Mockups
Appendices - D	Risk Analysis

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Once again, I express my gratitude to all those who have contributed to this project. Thank you

Introduction

Background and Context

Online recruitment fraud has emerged as one of the most pervasive forms of digital deception in the modern labor market. As job seekers increasingly rely on internet-based platforms to find employment, malicious actors have exploited these channels to post fraudulent job advertisements designed to harvest personal data, extort fees, or commit identity theft.

According to the Internet Crime Complaint Center (IC3), employment fraud caused over USD 214 million in losses in the United States alone in 2023. In the South Asian context, Sri Lanka has experienced a significant surge in fake recruitment advertisements, particularly targeting youth and recent graduates seeking employment. The Sri Lanka Police's Computer Crimes Investigation Division (CCID) reported a sharp increase in online employment scam complaints in recent years, with victims losing substantial sums to fake employers demanding upfront payments for non-existent positions.

The problem is compounded by a critical linguistic gap: the Sri Lankan workforce communicates predominantly in three languages — Sinhala, Tamil, and English. Yet every existing automated fake job detection system operates exclusively in English, leaving Sinhala- and Tamil-speaking job seekers entirely unprotected by current technology.

Furthermore, existing detection systems produce only a binary verdict (real or fake) without explaining their reasoning or providing users with an actionable risk level. This lack of transparency erodes user trust and provides no guidance on how serious the threat may be.

This research directly addresses SDG 8 (Decent Work and Economic Growth) by protecting workers from fraudulent employment schemes, and SDG 16 (Peace, Justice and Strong Institutions) by reducing digital crime and strengthening the integrity of digital labor market infrastructure in Sri Lanka.

Literature Review

A review of recent scholarly literature reveals significant progress in automated fake job detection, yet also exposes clear boundaries that current research has not crossed.

Dutta and Bandyopadhyay [1] conducted one of the earliest comprehensive evaluations using the EMSCAD Kaggle dataset (17,880 job postings), comparing Naive Bayes, K-NN, Decision Tree, Multi-Layer Perceptron, Random Forest, AdaBoost, and Gradient Boosting classifiers. Random Forest achieved 98.27% accuracy with a Cohen-Kappa score of 0.74, establishing a strong baseline. However, the study employs no natural language processing techniques beyond basic feature extraction and does not address the severe class imbalance inherent in the dataset.

Pillai [2] advanced this work by applying Bidirectional LSTM with word embeddings to the same dataset, achieving 98.71% accuracy and a ROC-AUC of 0.91. While this represents a notable improvement in sequential language modelling, the study lacks interpretability mechanisms and any form of risk stratification for end users.

Allam et al. [3] made the most thorough attempt at addressing class imbalance, evaluating six balancing strategies — including SMOTE-NC, Cluster-SMOTE, cost-sensitive learning, ensemble methods, GAN-based deep learning augmentation, and hybrid approaches — across four classifiers. Their best result (Random Forest with DL Augmentation) achieved precision of 98.6% and recall of 97.1%. However, the system remains a research prototype with no deployment interface and no explainability layer.

Pangare et al. [4] proposed JobScamShield, a notable system that integrates machine learning classification with external company verification via GST registry and Glassdoor API lookups. While architecturally innovative, the system is geographically constrained to India due to its dependence on the Indian GST registration system, and carries significant latency risk due to real-time external API calls at scale.

Singh et al. [5] introduced heuristic fraud signals such as free email domain detection (Gmail, Yahoo) and suspicious keyword identification ("urgent hiring", "no experience required"), achieving 94.2% accuracy with Logistic Regression. These engineered features represent useful domain knowledge but the system lacks deep semantic understanding from transformer-based models.

Sathwika et al. [6] deployed a Django web application integrating Random Forest, Naive Bayes, Logistic Regression, and SVM classifiers, with Logistic Regression achieving 98% accuracy. This represents one of the few production-facing deployments in the literature, yet the system remains English-only, static (no retraining pipeline), and provides no user-facing explanation.

Vijayalakshmi et al. [7] proposed an Elevated Learning strategy (ELFJP) — a custom deep learning ensemble — achieving 97.42% test accuracy. However, the ELFJP architecture is not fully described, constituting a black-box system with no explainability mechanism.

Devi et al. [8] compared supervised and unsupervised methods including Isolation Forest and Gaussian Mixture Models, achieving 98.27% with Random Forest. The study confirms ensemble methods as the dominant approach but does not explore NLP feature engineering or multilingual support.

In the domain of explainability, one recent study applied SHAP (Shapley Additive explanations) to XGBoost-based fraud detection for job postings, identifying top contributing features [9]. However, this study does not use transformer-based embeddings. Separately, XLM-RoBERTa has been fine-tuned for Sinhala text classification tasks including sentiment analysis and hate speech detection [10], demonstrating the viability of cross-lingual models for low-resource Sri Lankan languages — yet no study applies this capability to job fraud detection.

A review of existing products and services reveals that LinkedIn and Indeed implement proprietary fraud filters, but these are English-centric, opaque in their decision-making, and not tailored to South Asian job markets. No publicly available tool currently supports Tamil or Sinhala language job post analysis.

Research Gap

The review of existing literature exposes four critical and interconnected research gaps that this study addresses:

First, every published study on fake job detection — without exception — operates exclusively on English-language data. Sinhala and Tamil, the primary languages of Sri Lanka's workforce, remain entirely absent from the fake job detection research landscape. This represents a critical equity gap that leaves millions of job seekers without protection.

Second, all existing systems reduce detection to a binary classification (real or fake) or a raw probability score. No system provides a structured, tiered risk assessment that communicates to a lay user whether a posting is of low, medium, or high suspicion. This absence limits practical utility for users who need actionable guidance, not just a prediction.

Third, while BERT and transformer-based models have been applied to fake job detection, and while SHAP/LIME have been applied to other fraud detection tasks, no study combines transformer-based contextual embeddings with post-hoc explainability in the specific domain of fake job post detection. Users currently receive no explanation for why a posting is flagged.

Fourth, no publicly available multilingual fake job dataset exists for Tamil, Sinhala, and English drawn from real South Asian job portals. Constructing such a dataset — by scraping and manually labelling postings from live Sri Lankan platforms — is a prerequisite to advancing this field for South Asian markets and produces a more authentic and locally representative resource than translation-based approaches.

This research fills all four gaps simultaneously: it is the first to (i) support Tamil, Sinhala, and English; (ii) provide a three-tier Suspicion Scoring Mechanism; (iii) integrate SHAP/LIME explainability with a transformer model in this domain; and (iv) produce a trilingual labelled dataset scraped from real Sri Lankan job portals as a reusable research artefact.

Objectives

Main Objective

To design, develop, and evaluate a multilingual AI-powered mobile application that detects fake job postings in Sinhala, Tamil, and English using XLM-RoBERTa, assigns a three-tier Suspicion Score (Low / Medium / High), and provides SHAP/LIME-based natural language explanations — achieving a minimum F1-score of 90% across all three languages by the end of the academic year.

Specific Objective

1. To construct a Sri Lanka-specific trilingual fake job posting dataset by scraping job advertisements from topjobs.lk and ikman.lk using automated web scraping tools (Scrapy and Selenium), collecting fields including job title, company name, description, requirements, salary, and contact email, and manually labelling each posting as fake or legitimate through a structured annotation process by trained reviewers, achieving a minimum inter-annotator agreement of Cohen's Kappa ≥ 0.80 across a validation sample.
2. To fine-tune the XLM-RoBERTa cross-lingual transformer model on the trilingual dataset, applying SMOTE-based class balancing to address the class imbalance observed in the collected dataset, and achieve a minimum classification F1-score of 90% across English, Sinhala, and Tamil test sets.
3. To design and implement a three-tier Suspicion Scoring Mechanism that computes a weighted composite score from classifier probability, heuristic fraud signals (keyword presence, free email domain, missing structural fields), and structural metadata — mapping scores to Low, Medium, and High risk levels with defined score thresholds.
4. To integrate SHAP and LIME post-hoc explainability frameworks to generate human-readable, per-posting explanations identifying the top contributing features to the suspicion classification, validated through a user comprehension study with at least 30 participants.
5. To develop and deploy a cross-platform mobile application (Android and iOS using React Native) exposing the detection system via a RESTful API backend, supporting all three languages in the user interface.

6. To evaluate the complete system end-to-end on a held-out test set and through a user study measuring perceived trustworthiness, usefulness, and comprehension of explanations using a Likert-scale questionnaire administered to a minimum of 50 participants.

Methodology

Research Approach

This research adopts a Design Science Research (DSR) methodology, which is appropriate for studies that produce and evaluate an IT artefact addressing a real-world problem. The research proceeds through five iterative phases: problem identification, solution design, artefact development, evaluation, and communication.

Technical Methods and Algorithms

Dataset Construction

Job postings are collected directly from two major Sri Lankan job portals — topjobs.lk and ikman.lk — using automated web scraping built with Scrapy and Selenium. The scraper collects publicly visible posting fields including job title, company name, description, requirements, salary range, and contact email address, while respecting each site's robots.txt rules and applying request rate limiting to avoid server burden. Postings are collected across English, Sinhala, and Tamil languages as they naturally appear on these platforms. Each collected posting is then manually labelled as fake or legitimate by trained annotators following a structured annotation guideline. A minimum inter-annotator agreement of Cohen's Kappa ≥ 0.80 is required across a random validation sample before the labelled data is accepted for model training. The target dataset size is approximately 5,000–10,000 labelled postings across the three languages.

Class Imbalance Handling

The scraped and labelled dataset is expected to exhibit class imbalance between real and fake postings. The actual ratio will be determined after the labelling phase is complete. SMOTE-ENN (Synthetic Minority Oversampling Technique combined with Edited Nearest Neighbors) is applied during training to synthesize minority class examples and clean borderline majority class samples.

Model Architecture

XLM-RoBERTa (xlm-roberta-base) — a cross-lingual transformer model pre-trained on 100 languages including Sinhala and Tamil — is fine-tuned using the Hugging Face Transformers library. The model input comprises concatenated job posting fields (title + company profile + description + requirements + benefits). A classification head (linear layer + SoftMax) is attached for binary prediction. Training uses AdamW optimizer with a learning rate of 2e-5, batch size of 16, and early stopping with patience of 3 epochs.

Suspicion Scoring Mechanism

The Suspicion Score is computed as a weighted composite:

Suspicion Score = $0.50 \times P(\text{fake}) + 0.20 \times \text{Keyword Score} + 0.15 \times \text{Structural Penalty} + 0.15 \times \text{Domain Score}$

- P(fake): Model's predicted probability of fraudulence (XLM-RoBERTa output)
- Keyword Score: Normalized count of high-risk phrases (e.g., 'urgent hiring', 'no experience needed', 'work from home — earn daily')
- Structural Penalty: Proportion of critical fields (company profile, salary, requirements) that are empty or implausibly short
- Domain Score: Binary flag (1 = free email domain such as Gmail or Yahoo; 0 = corporate domain)

Score thresholds: Low Risk = 0.00–0.39 | Medium Risk = 0.40–0.69 | High Risk = 0.70–1.00

Explainability Layer

SHAP (Shapley Additive explanations) is applied to generate global feature importance rankings across the test set. LIME (Local Interpretable Model-agnostic Explanations) is applied per-prediction to highlight the top five words and features contributing to the classification. Both outputs are converted to natural language explanation templates displayed within the mobile application.

Tools and Platforms

Table 2 Tools and Platforms

Tools / Platform	Purpose	Justification
Python 3.10	ML development	Industry standard for NLP/ML
Hugging Face Transformers	XLM-RoBERTa fine-tuning	Best library for transformer models
SHAP / LIME libraries	Post-hoc explainability	State-of-the-art XAI frameworks
React Native	Mobile app (iOS & Android)	Cross-platform, single codebase
FastAPI	REST API backend	High-performance Python API framework
Scrapy / Selenium	Web scraping (topjobs.lk & ikman.lk)	Industry-standard Python scraping frameworks
AWS EC2 / Lambda	Model serving & deployment	Scalable cloud infrastructure
MongoDB Atlas	Data storage	Schema-flexible NoSQL for varied job structures
GitHub Actions	CI/CD pipeline	Automated testing and deployment

Data Collection and Ethical Considerations

Job posting data is collected from publicly visible pages on topjobs.lk and ikman.lk using automated web scraping. Scraping is conducted ethically: the scraper respects each site's robots.txt directives, applies request rate limiting to avoid placing undue load on the servers, and complies with the sites' terms of service. No recruiter personal data beyond what is publicly displayed in the job advertisement is collected or stored. All data is used solely for academic research purposes.

A user study will be conducted following ethical approval from the SLIIT Research Ethics Committee. Participants will provide informed consent, participation will be voluntary and anonymous, and no personal data beyond Likert-scale responses will be recorded. Survey instruments are included in Appendix A.

Validation Strategy and Performance Metrics

Model performance is evaluated using a stratified 80/20 train-test split with 5-fold cross-validation. Primary metrics include Precision, Recall, F1-Score, and ROC-AUC. The class-weighted F1-score is the primary optimization target due to class imbalance. Dataset labelling quality is measured using Cohen's Kappa across a random validation sample of annotated postings, with a minimum threshold of ≥ 0.80 required before the dataset is accepted for training. Explainability quality is assessed through a user comprehension study using a Likert-scale instrument (1–5). System performance is measured by API response latency (target: < 2 seconds per query).

High-Level System Architecture

Overview of the Proposed Solution and Its Components

The proposed system is a multilingual AI-powered fake job post detection system deployed as a cross-platform mobile application supporting English, Sinhala, and Tamil. The system accepts both text and image-based job postings, analyses them through a transformer-based classification pipeline, and returns a risk-tiered suspicion score with a human-readable explanation — all within a single mobile interface.

The solution is structured across four layers. The **Presentation Layer** comprises a React Native mobile application providing the user-facing interface including language selection, job post submission, suspicion score display, SHAP/LIME explanation view, and safety checklist. The **Backend Intelligence Layer** hosts the core processing pipeline — Tesseract OCR, language detection, XLM-RoBERTa inference, heuristic signal extraction, Fuzzy Logic suspicion scoring, and SHAP/LIME explainability — all orchestrated through a FastAPI REST API deployed on AWS. The **Data Layer** consists of MongoDB Atlas for prediction and feedback storage and AWS S3 for versioned model weight management. The **Security Layer** enforces JWT authentication, TLS 1.3 encryption, and input sanitisation across all API endpoints.

Together these four layers form a cohesive, scalable, and explainable fake job detection system specifically designed for the Sri Lankan multilingual job market.

Data Flow

A user submits a job posting (text or URL) via the mobile application. The backend automatically detects the language and routes the text through the XLM-RoBERTa inference engine. Simultaneously, the Heuristic Signal Extractor evaluates keyword patterns, structural completeness, and email domain type. The Suspicion Score Calculator combines these outputs into a weighted composite score mapped to a Low/Medium/High risk tier. The SHAP/LIME engine generates a per-prediction explanation identifying the top contributing features. All results are returned to the mobile application in under two seconds.

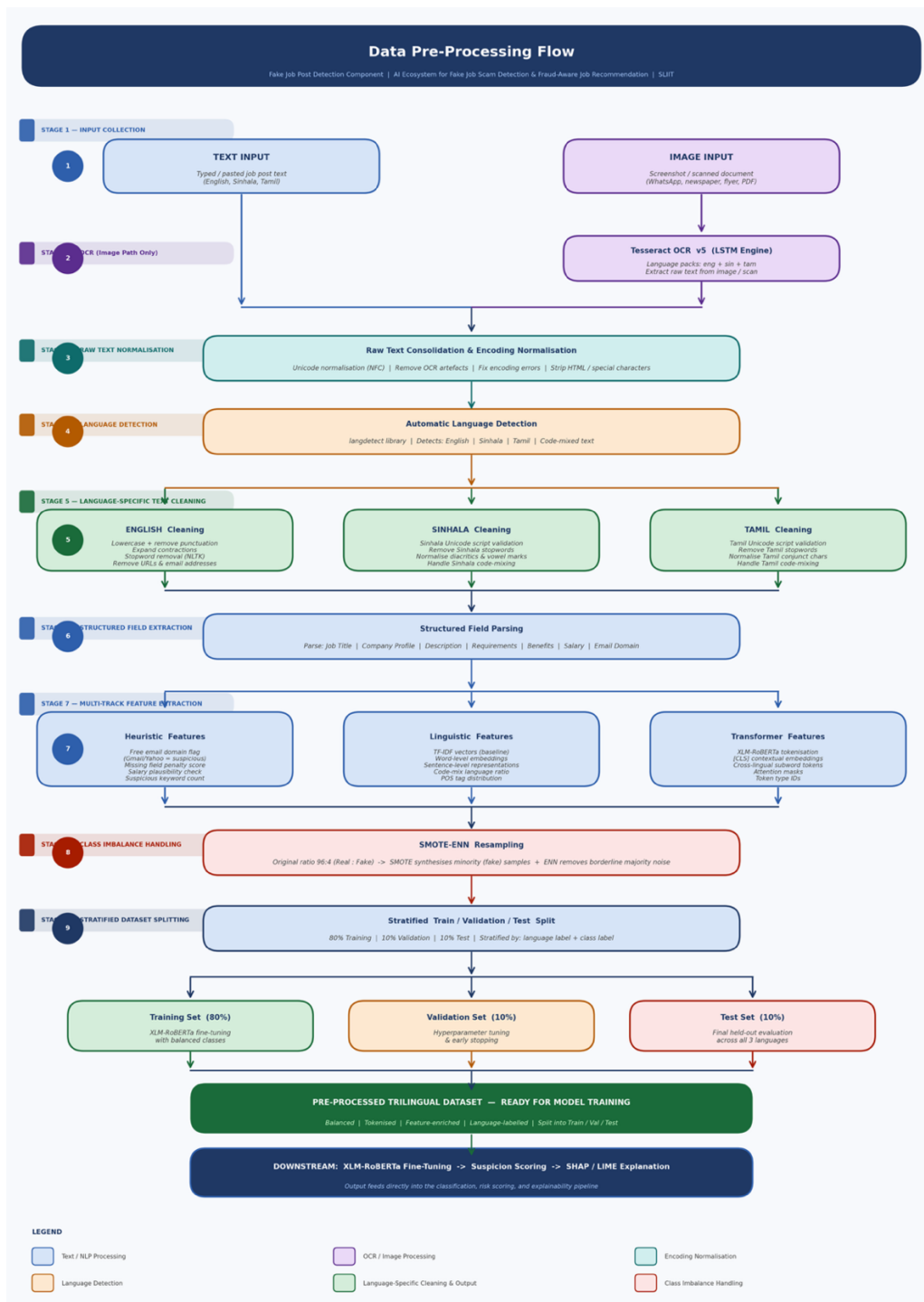


Figure 1 Data Flow

System Diagram of the Component

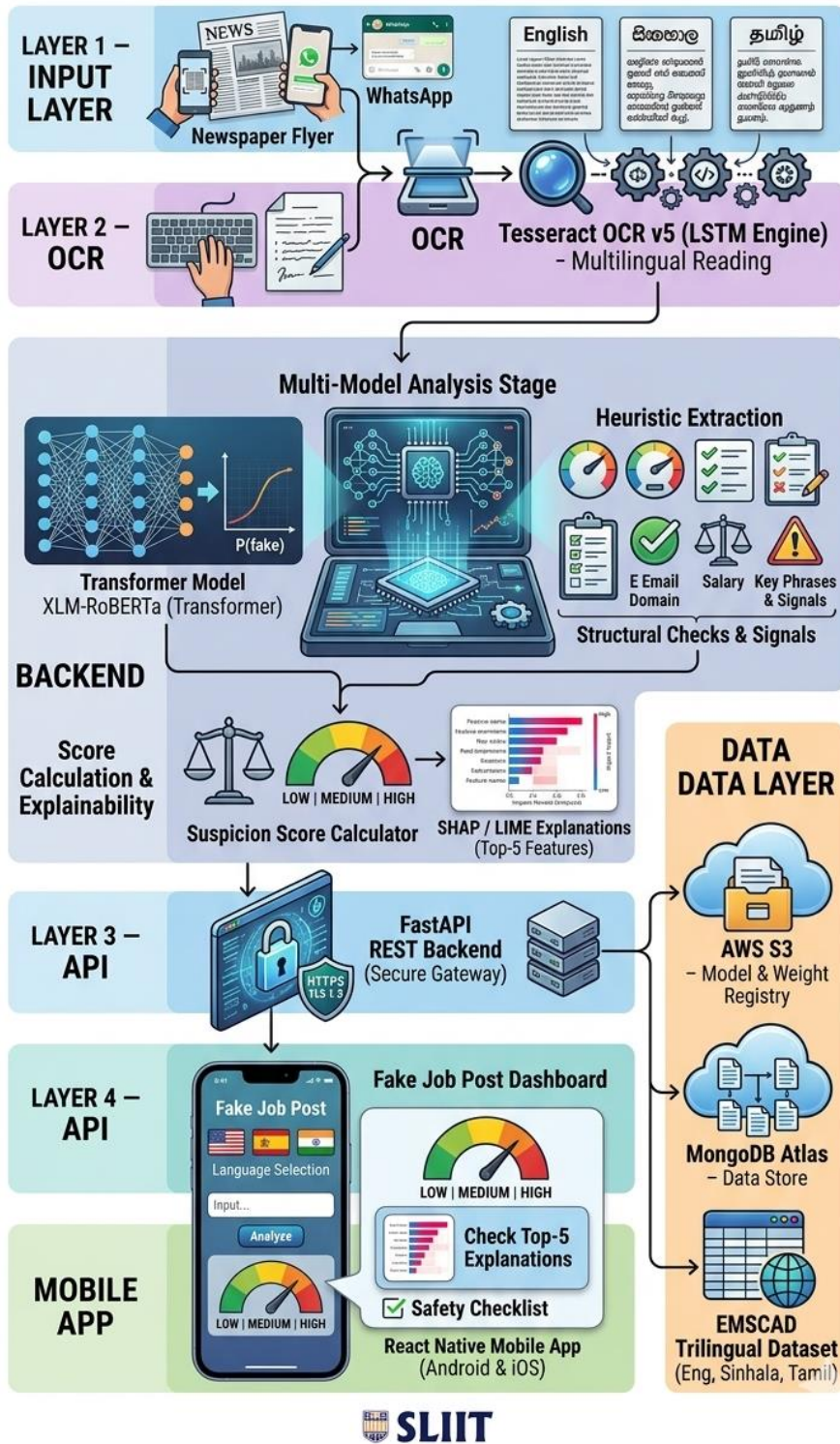


Figure 2 System Diagram of the Component

Component Interactions

The six core components of the system interact in a sequential pipeline coordinated by the FastAPI REST API. The **Mobile Application** dispatches user input — text or image — to the API via authenticated HTTPS requests. For image inputs, the **Tesseract OCR Engine** first extracts raw text before passing it downstream. The **Language Detection Module** then identifies the language (English, Sinhala, Tamil, or code-mixed) to govern tokenization.

The **XLM-RoBERTa Inference Engine** and the **Heuristic Signal Extractor** operate in parallel — the former producing a fraud probability $P(\text{fake})$, the latter extracting keyword density, structural completeness, and domain credibility signals. These four inputs are fed into the **Fuzzy Logic Suspicion Score Calculator**, which applies Mamdani inference rules and Centroid of Area defuzzification to produce a crisp risk score mapped to Low, Medium, or High tier.

The **SHAP/LIME Explainability Engine** then generates a per-prediction natural language explanation identifying the top five contributing features, returned to the mobile application alongside the score and verdict. All prediction records are written asynchronously to **MongoDB Atlas**, while model weights are fetched from and versioned on **AWS S3**.

Scalability Considerations

The backend is deployed as containerized microservices on AWS using Docker and orchestrated with AWS ECS. The XLM-RoBERTa model is served via AWS Lambda with auto-scaling policies to handle variable request volumes. MongoDB Atlas provides horizontal sharding for the data layer. API Gateway rate limiting prevents abuse.

Security Considerations

All API endpoints require JWT-based authentication. Data in transit is encrypted via TLS 1.3. No personally identifying information is stored server-side. User-submitted job postings are hashed before storage for duplicate detection. Input validation and sanitization are applied at the API gateway layer to prevent injection attacks.

User Requirements

Stakeholder Identification

Table 3 Stakeholder Identification

Stakeholder	Role	Primary Need
Job Seekers	End users submitting job posts for analysis	Understand risk level and explanation in their language
System Administrators	Manage model updates, user reports, and system health	Dashboard for monitoring predictions and flagged posts
Researchers / Academics	Access to model performance data and dataset	Reproducible results and open dataset access
Sri Lankan Job Portals	Potential integration partners via API	Automated pre-screening of posted jobs

Functional Requirements

1. The system shall accept a job posting as free-text input or URL in English, Sinhala, or Tamil and automatically detect the input language.
2. The system shall classify each job posting as Real or Fake using the fine-tuned XLM-RoBERTa model with a minimum confidence threshold of 60%.
3. The system shall compute and display a Suspicion Score on a 0–1 scale and map it to a colour-coded risk tier: Low (green), Medium (amber), or High (red).
4. The system shall generate a SHAP/LIME-based explanation in the user's language identifying the top five features contributing to the prediction.
5. The system shall allow users to report false positives or false negatives, storing feedback for model retraining cycles.
6. The system shall provide a job safety checklist based on the suspicion score, advising users on precautions for Medium and High risk posts.
7. The system shall maintain a personal history log of all job posts a user has submitted for analysis, accessible within the mobile application.
8. The system shall support English, Sinhala, and Tamil in the complete user interface, including all labels, explanations, risk descriptions, and alerts.

Non-Functional Requirements

Security

All user data must be transmitted over TLS 1.3. JWT tokens must expire within 24 hours. No personally identifiable information shall be stored without explicit consent.

Performance

The system shall return a complete prediction, suspicion score, and explanation within 2 seconds for 95% of requests under normal load (up to 500 concurrent users).

Usability

The mobile application shall achieve a System Usability Scale (SUS) score of at least 70 in user testing across all three language groups.

Scalability

The backend architecture shall support horizontal scaling to accommodate a tenfold increase in request volume without architectural changes.

Reliability

The system shall achieve 99.5% uptime measured monthly, with automated health checks and failover mechanisms in place.

Maintainability

Model weights and application code shall be versioned using semantic versioning. A retraining pipeline shall be triggered automatically when user feedback accumulates beyond 500 new labelled samples.

Commercialization Plan

Target Market and Customer Persona

Primary market: Sri Lankan job seekers aged 18–35, particularly recent graduates and those actively using online job portals. Secondary market: Sri Lankan recruitment agencies and job portals (TopJobs.lk, XpressJobs.lk) seeking to improve platform integrity. Tertiary market: South Asian markets where Tamil is widely spoken (India — Tamil Nadu, Singapore, Malaysia).

Value Proposition

The proposed system is the only fake job detection tool that speaks your language — literally. Unlike global fraud filters that operate only in English, the system analyses job postings in Sinhala, Tamil, and English, tells you exactly why a posting is suspicious, and gives you a clear Low/Medium/High risk rating so you know whether to apply, proceed cautiously, or walk away.

Revenue Model

- Freemium Mobile App: Free tier (5 analyses/day); Premium subscription at LKR 299/month (unlimited analyses, history, priority processing)
- B2B API Licensing: Job portal integration via API at LKR 15,000/month for up to 10,000 API calls
- Enterprise Partnership: Custom deployment for large recruitment agencies or government employment portals

Cost Estimation

- AWS EC2 / Lambda (model serving): LKR 8,000/month
- MongoDB Atlas (data storage): LKR 3,000/month
- Scraping infrastructure & proxy service (ongoing): LKR 2,000/month
- App Store / Play Store developer accounts: LKR 15,000 one-time

Competitive Advantage

No existing product — commercial or open-source — offers fake job detection in Sinhala or Tamil. This linguistic specificity, combined with transparent AI explanations and risk-tiered scoring, creates a defensible differentiation in the Sri Lankan market that cannot be replicated by global platforms without significant localisation investment.

Intellectual Property Considerations

The trilingual fake job posting dataset scraped and labelled during this research will be published under a Creative Commons Attribution licence to advance the research community. The Suspicion

Scoring algorithm and XLM-RoBERTa fine-tuning pipeline will be considered for provisional patent filing through SLIIT's technology transfer office upon completion of the research.

Budget and Justification

Table 4 Budget and Justification

Item	Cost	Justification
AWS EC2	20,000	Required for XLM-RoBERTa inference at scale
Scraping Infrastructure & Proxy Service	8,000	Web scraping of topjobs.lk & ikman.lk for dataset construction
MongoDB Atlas (12 months)	18,000	Cloud database for job post and feedback storage
Google Play Developer Account	9,000	Required for Android Play Store deployment
Research Materials & Printing	7,000	Survey instruments, interim and final reports
	62,000	

Work Breakdown Structure

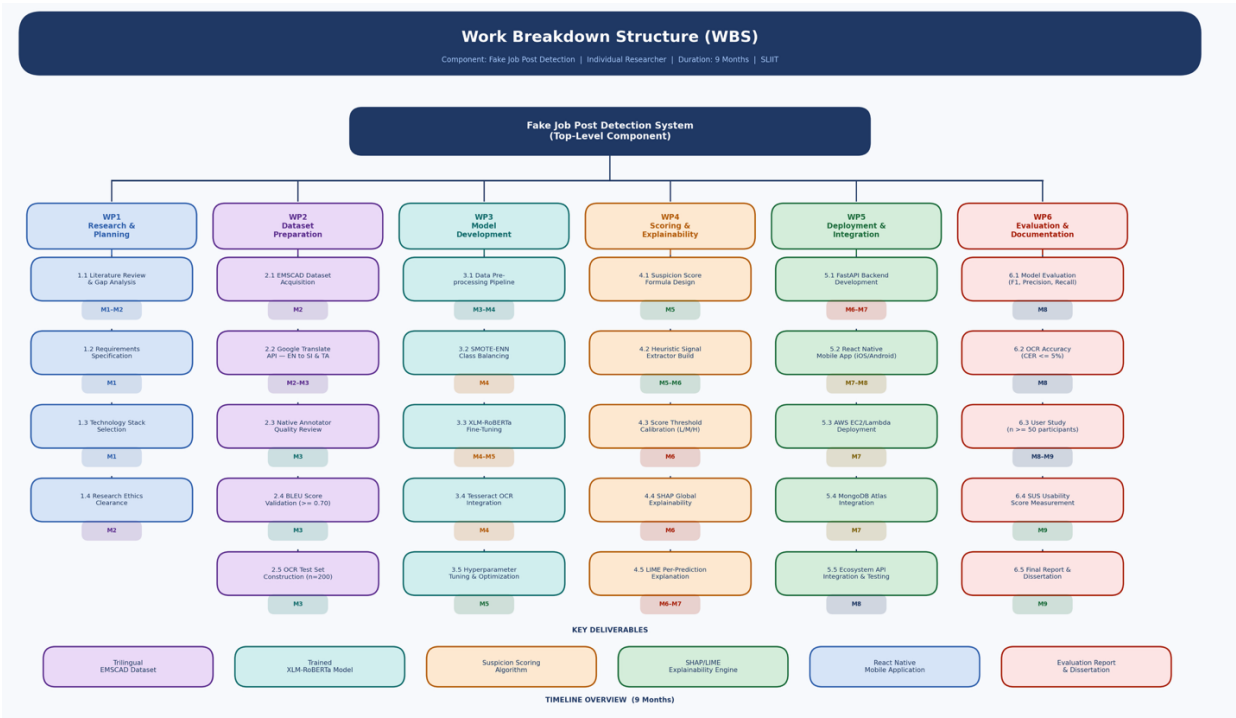


Figure 3 Work Break Down Structure

Gantt Chart

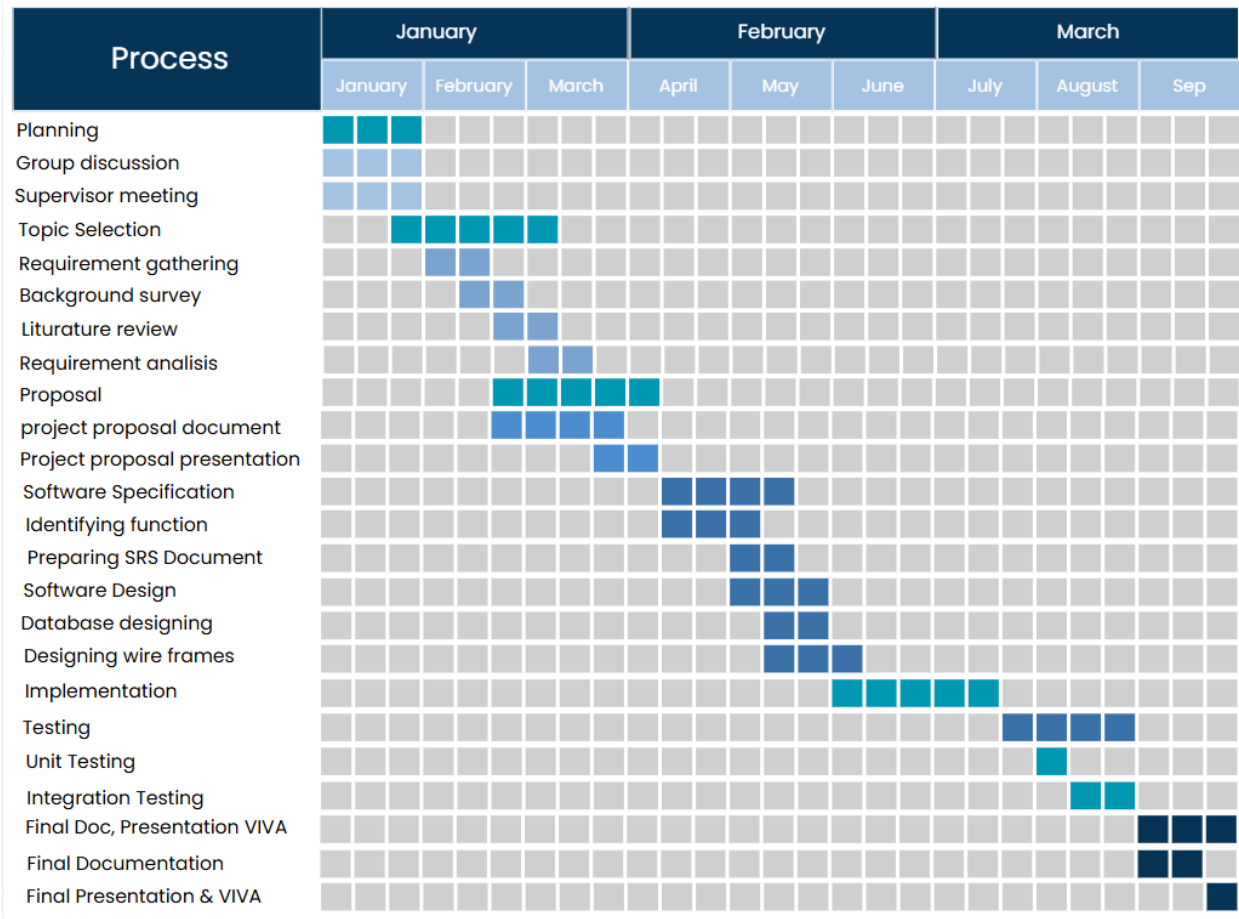


Figure 4 Gantt Chart

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Appendices

Appendices – A

This survey was administered via Google Forms between 5 March 2026 and 9 March 2026 as part of the preliminary needs assessment for this research. The purpose of the survey was to assess participant awareness of online job scams, experience with fake job postings, and attitudes toward AI-based detection tools. A total of 57 valid responses were collected from job seekers, undergraduate students, and working professionals in Sri Lanka. Participation was voluntary and all responses were collected anonymously.

Survey Title: Online Job Scam Awareness & AI Detection System Survey

Target Audience: Job seekers, undergraduate students, and working professionals in Sri Lanka

Total Respondents: 57

Distribution Period: 5 March 2026 – 9 March 2026

Distribution Method: Google Forms (online link shared via WhatsApp)

Section 1 — Demographic Information

Q1. What is your age group? 18–22 / 23–27 / 28–35 / Above 35

Q2. What is your current status? Undergraduate student / Job seeker / Working professional / Graduate

Q3. Have you ever searched for jobs online? Yes / No

Section 2 — Online Job Search Behaviour

Q4. Which platforms do you use to search for jobs? LinkedIn / Job websites (TopJobs, Indeed, etc.) / WhatsApp groups / Email invitations / Facebook job groups / Telegram / Other

Q5. Have you ever seen fake or suspicious job advertisements online? Yes / No / Not sure

Q6. What characteristics make a job advertisement look suspicious? Unrealistically high salary / Very little experience required / Lack of company information / Urgent hiring messages / Poorly written job descriptions

Section 3 — Recruiter Communication Experience

Q7. Do you usually check the company details before applying for a job? Yes / No / Sometimes

Q8. Which company-related factors increase your trust in a job opportunity? Official company website / Corporate email address / Contact information / Social media presence / Company reputation

Q9. Have you ever received suspicious messages from someone claiming to be a recruiter? Yes / No / Not sure

Q10. Through which platforms did you receive suspicious recruiter messages? WhatsApp / Email / LinkedIn / Facebook / Telegram / Other / I have never received suspicious recruiter messages

Q11. What type of suspicious behavior did you notice in those messages? Asking for money or a processing fee / Asking for personal documents / Urgent pressure to respond quickly / Suspicious links / Fake company name / Unrealistic salary offers

Q12. Have you ever been asked to pay money for a job opportunity? Yes / No / Not sure

Section 4 — System Need and Feature Preferences

Q13. How confident are you in identifying fake job offers or recruitment scams? (Scale: 1 = Not confident at all — 5 = Very confident)

Q14. Do you think recruitment scams are increasing on online platforms? Yes / No / Not sure

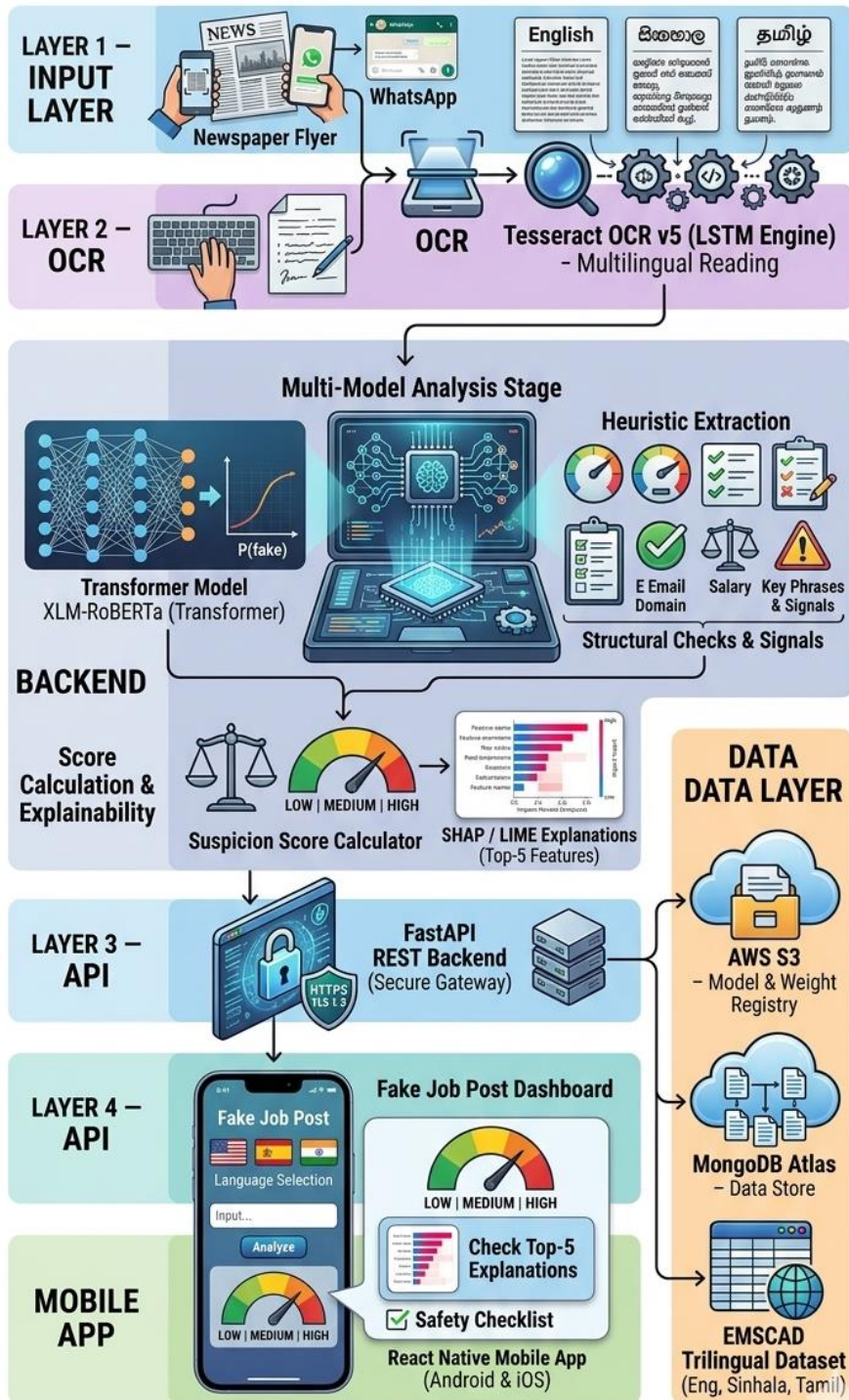
Q15. Would you use a system that automatically detects scam job messages? Yes / No / Maybe

Q16. Which features would help you detect job scams more effectively? Scam risk score for messages / Explanation of why a message is suspicious / Warning for suspicious links / Verification of recruiter details / Alerts for possible scam messages

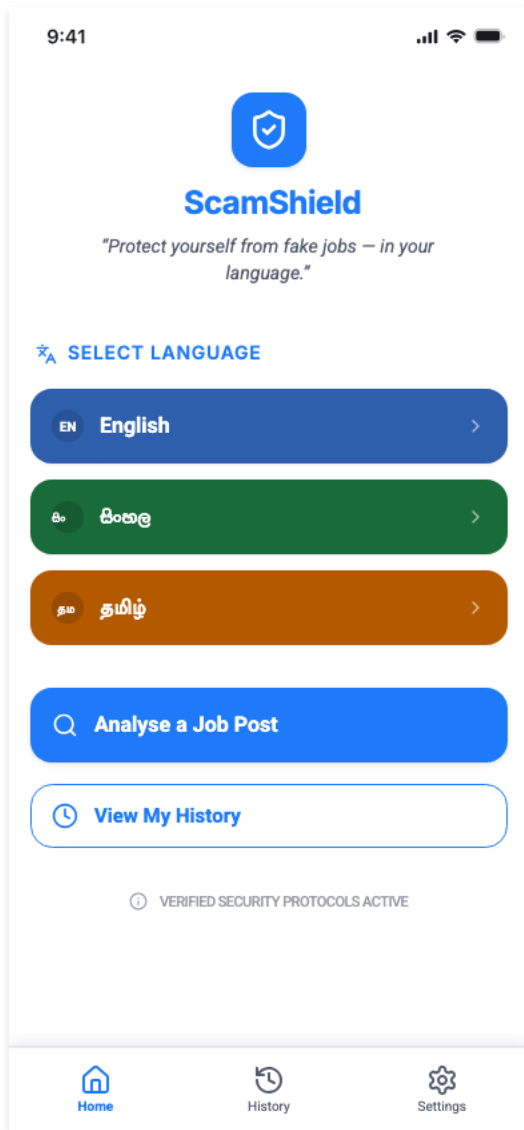
Q17. Do you think AI technology can help reduce recruitment scams and protect job seekers? Yes / No / Not sure

Q18. In your opinion, what is the biggest risk when applying for jobs online? (Open-ended)

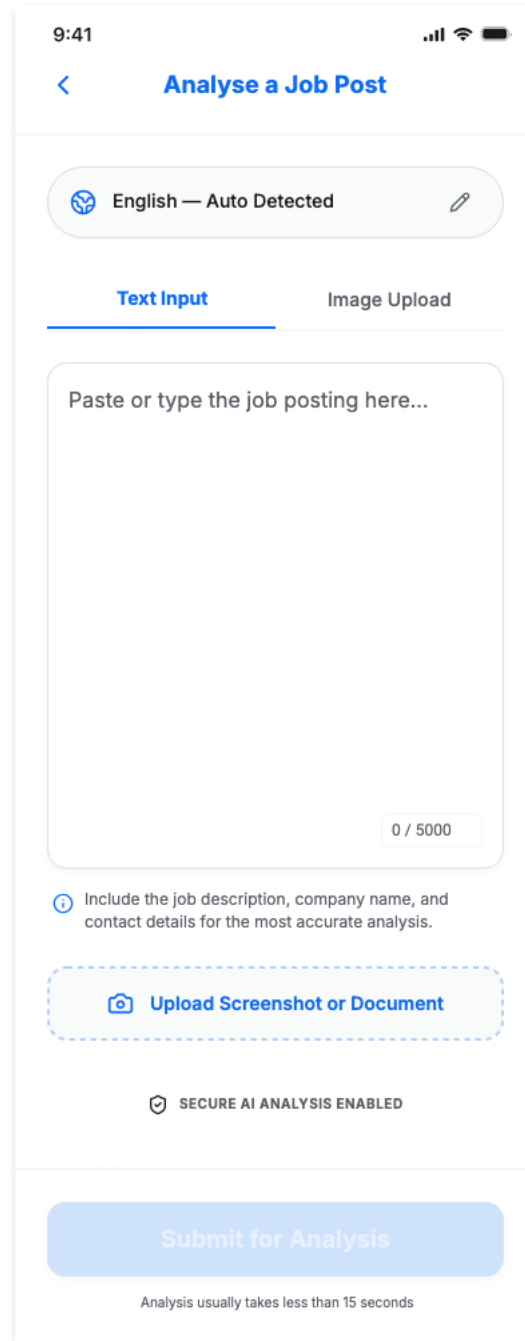
Appendices - B



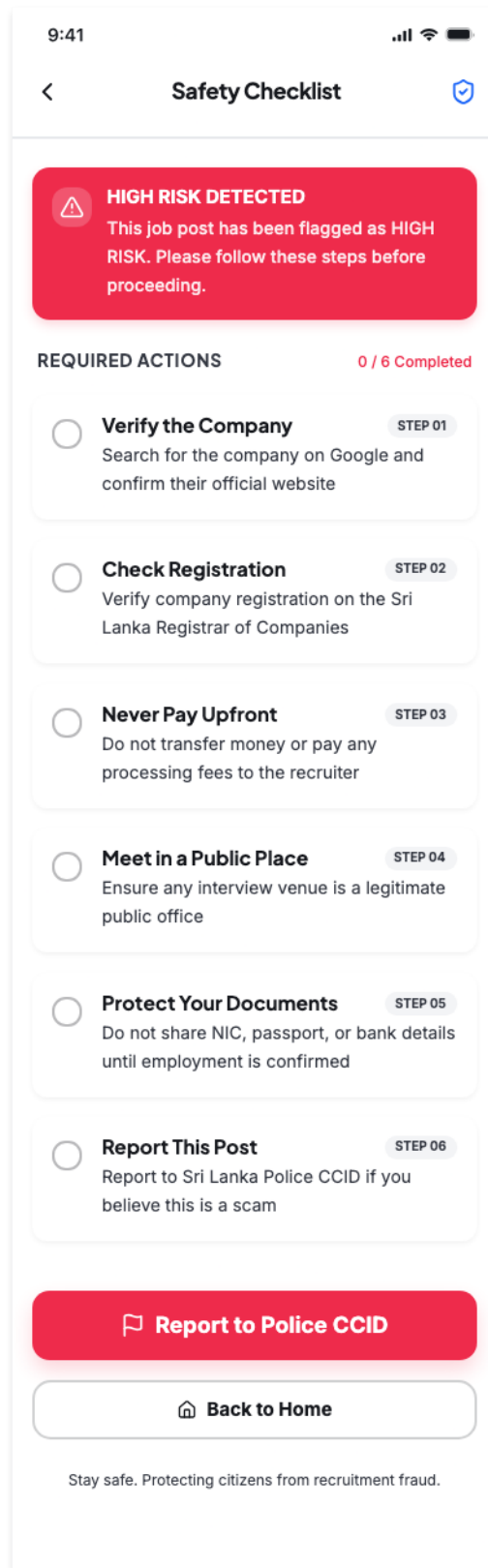
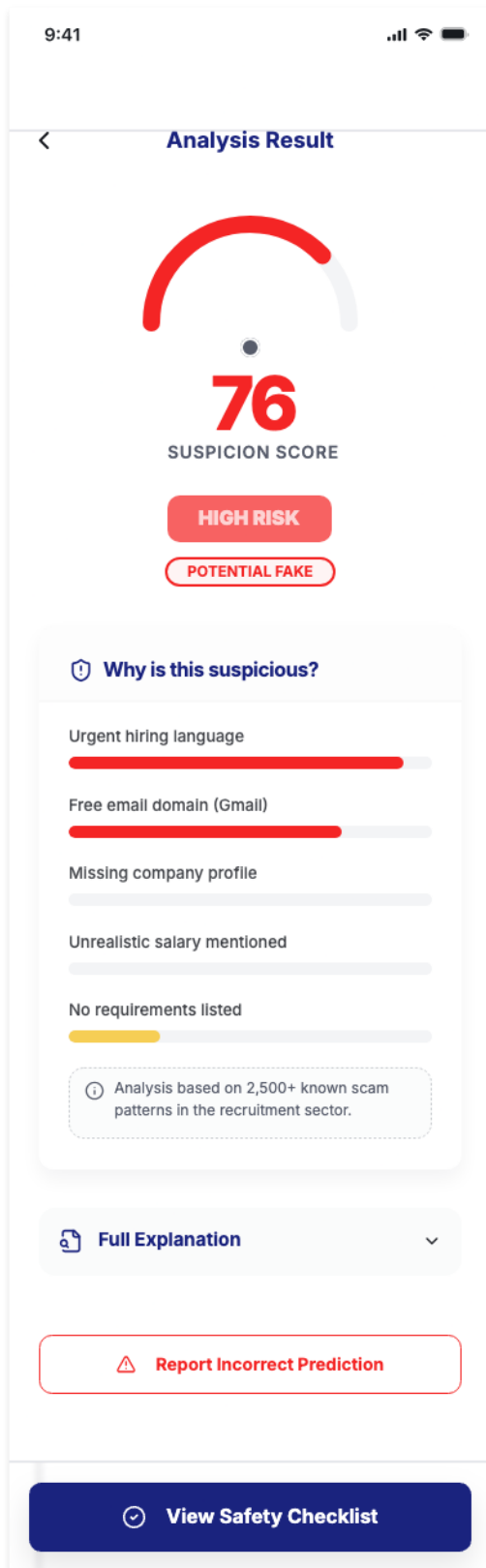
Appendices – C



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Appendices – D Risk Analysis

Table 5 Risk Analysis

ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R1	Annotator labelling inconsistency producing low inter-annotator agreement	Medium	High	Set Cohen's Kappa ≥ 0.80 as the labelling quality gate; provide detailed annotation guidelines with worked examples; conduct calibration sessions before full labelling begins; re-adjudicate disagreed samples
R2	XLM-RoBERTa F1-score below 90% on non-English sets	Medium	High	Data augmentation; increase annotated sample size; hyperparameter tuning; try larger model variant
R3	Tesseract OCR accuracy below target CER $\leq 5\%$	Medium	Medium	Pre-process images (denoise, binarise, deskew); test multiple Tesseract PSM modes; supplement with Google Vision API
R4	AWS cost overruns exceeding budget	Low	Medium	Set AWS Budgets alerts at 80% threshold; use Lambda with concurrency caps; monitor monthly via Cost Explorer
R5	Insufficient user study participants (target $n \geq 50$)	Low	Medium	Recruit via SLIIT student portal, community networks, and WhatsApp; offer incentives; begin recruitment early
R6	App Store / Play Store rejection	Low	High	Review Apple and Google content policies before submission; ensure no data collected without explicit user consent
R7	Model bias against legitimate informal job postings	Medium	Medium	Threshold calibration; adversarial test cases; user feedback loop for false positive correction
R8	Fuzzy logic rule base produces inconsistent risk tiers	Low	Medium	Validate all 18 rules against manually labelled test cases; cross-check outputs with domain expert review
R9	Scraped job postings contain incomplete or corrupted fields due to dynamic page rendering or	Medium	High	Use Selenium for JavaScript-heavy pages; implement field-level validation to flag posts with missing critical fields; schedule periodic scraper maintenance checks; maintain fallback CSS selectors for common layout changes

	site structure changes			
R10	Project timeline slippage due to technical complexity	Medium	Medium	Build 2-week buffers into each WBS milestone; weekly supervisor check-ins; priorities core MVP features first